

Inspiration

Biomimicry is about learning from nature's shapes, movements, and systems to spark new design ideas. **Animal courtship displays** are nature's own performances to grab attention, send signals, and stir emotion.



These courtship displays can inspire expressive and engaging ways to design data visualizations.

Color

Form & Motion

Inspire Design in InfoVis

Metaphor & Comparison Animation Path

Easing Curve Rhythm

Flirting Charts

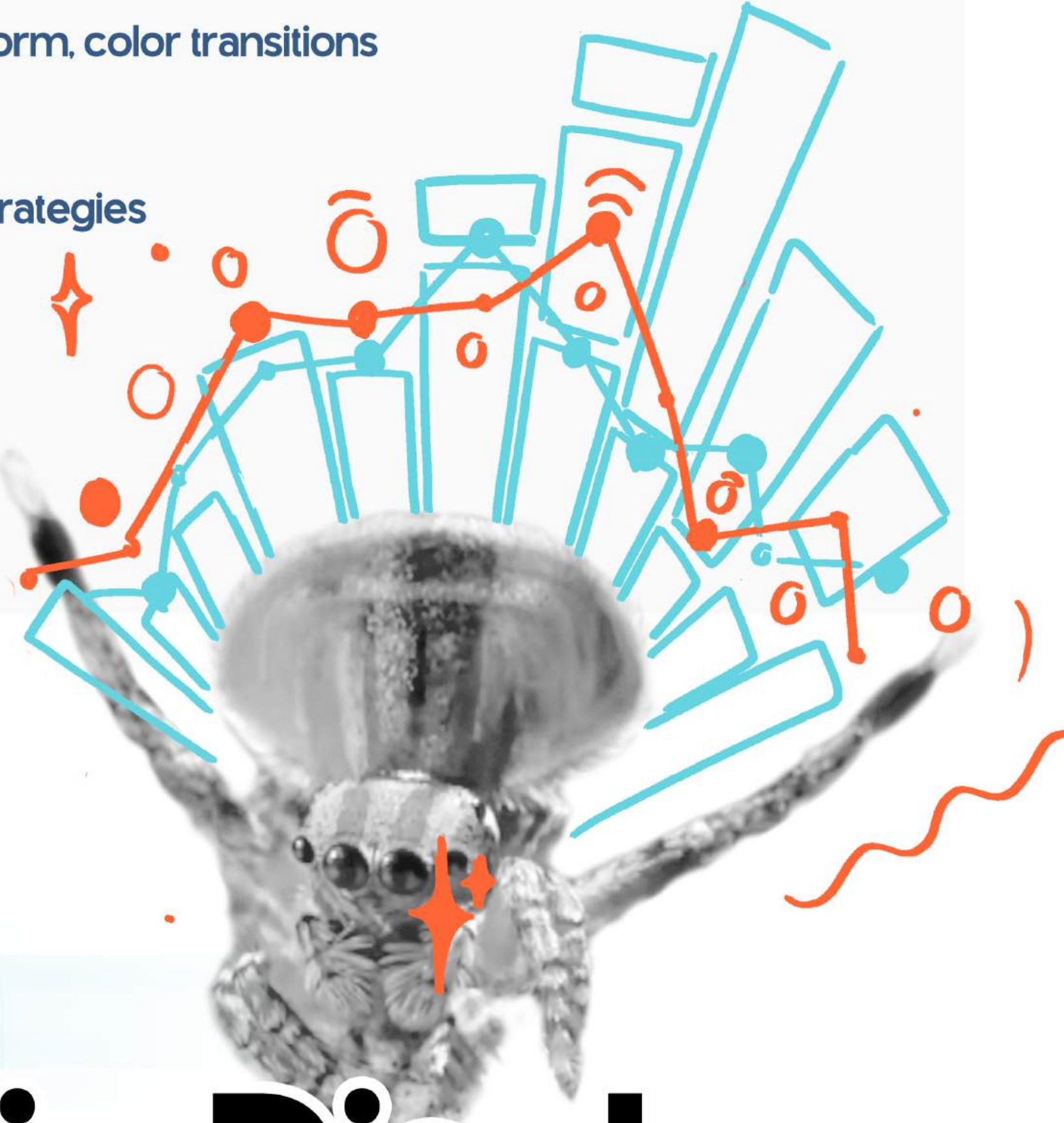
Expressive Motion Design Inspired by Animal Courtship Displays

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Method

To explore how animal courtship performances can inspire expressive InfoVis, we conducted a **qualitative coding** of video-recorded courtship displays.

- 1 Collected high quality animal courtship videos and research studies
- 2 Selected example species: Peacock Spider, Bird-of-Paradise, Cuttlefish
- 3 Qualitative coding of motion, form, color transitions
- 4 Mapped to InfoVis animation strategies



Peacock Spider

opisthosomal fan open

legs wave

Radial unfolding with dynamic edges to attract peripheral attention

Cuttlefish

zebra pattern to light mottle light mottle to uniform darkening

Transition between lines and dots

Bird of Paradise

move head wires sudden flash turn head side to side

Map data to angles or wire lengths Attention flash Peripheral attention dots

Walk and pause dance display

This work opens several directions for future research.

Empirical Studies: Assess whether courtship-inspired motion improves engagement, recall, and emotional resonance in InfoVis.

Formalization: Formalize natural motion rhythms and dynamics into coded patterns for broader application.

New Mediums: Apply these motion codes to data physicalization and XR using materials like soft robotics and motion fabrics to create immersive data experiences.

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